INTRODUCTION

The integrity of democratic elections is vital for legitimate governance, yet it remains vulnerable to systemic issues like vote-buying, physical ballot manipulation, and technical malfunctions (Tsek.ph, 2022). In the Philippines, skepticism surrounding automated election systems is often exacerbated by machine failures and the persistent threat of identity fraud or "ghost voting" (Eligo, 2024). Traditional systems typically rely on centralized databases, which present a significant security risk by creating a single point of failure susceptible to administrative tampering or external cyber-attacks (Khan et al., 2018).

To address these vulnerabilities, the **VOTECHAIN** system is proposed as a decentralized, IoT-based solution that leverages blockchain technology and multi-factor biometric authentication. This approach aims to restore public confidence by providing a verifiable and transparent electoral process that can be audited by any citizen in real-time.

Overview of the Project

The **VOTECHAIN (Voting Optimization Through Encrypted Blockchain)** system is grounded in the fundamental principles of embedded systems, biometrics, and distributed ledger technology, operating through a synchronized loop of authentication and immutable recording. The core concept begins with **Multi-Factor Authentication (MFA)**, where the system utilizes an **AS608 Fingerprint Sensor** and **RFID technology** to verify unique physical and digital identifiers before granting ballot access.

Once identity is confirmed, the **Embedded Processing** layer—managed by a **Raspberry Pi 5**—handles the voter's selections and prepares the data for the blockchain. The system then applies the principle of **Cryptographic Hashing (SHA-256)** to ensure data integrity; each vote is converted into a unique hash that links to the previous block, creating a permanent chain that cannot be altered without detection (Khan et al., 2018). Finally, the system employs **Internet of Things (IoT) Connectivity** to stream these encrypted transactions to a self-hosted **public verification web app**, allowing for real-time result visualization and enabling voters to cross-reference their personal transaction hashes against the public record for absolute transparency.

Objectives of the Project

This project aims to design and develop a secure, integrated electronic voting system that combines two-factor hardware authentication with a private, permissioned blockchain backend to ensure data immutability, prevent double-voting, and facilitate real-time public auditing. Specifically, the students aim to:

- Integrate biometric (fingerprint) and RFID modules into a voting kiosk that communicates with a central API to enforce a strict one-to-one mapping between physical credentials and voter identities, preventing credential sharing.
- Implement a Python-based blockchain engine using SHA-256 hashing algorithms that instantly seals submitted ballots into a linked list structure, ensuring that any retroactive data modification triggers a detectable validation failure.
- Deploy a WebSocket-enabled frontend using Django Channels that broadcasts vote tallies and blockchain updates to public observers instantly, while providing an interactive tool for independent verification of the cryptographic chain integrity.
- Validate the system's end-to-end reliability by testing the data pipeline from the physical sensor input at the edge to the permanent, immutable storage in the central ledger.

Scopes and Limitations

The study focuses on the development of a precinct-level electronic voting system that integrates a custom-built, private blockchain backend with physical hardware authentication. The scope is limited to a permissioned blockchain architecture where a central election server acts as the sole authority for validating and committing blocks, utilizing database transaction locks rather than decentralized consensus to ensure data consistency. Consequently, the system does not employ Proof-of-Work or difficulty puzzles, prioritizing immediate transaction finality and energy efficiency over the trustless distribution found in public cryptocurrencies. Operational functionality is dependent on a continuous network connection between the voting terminal and the central server to perform real-time API credential checks and ballot submission, meaning the current iteration does not support fully offline, asynchronous voting.

REVIEW OF RELATED LITERATURE

This section explores related literature about electoral integrity, biometric security, and decentralized ledger technology that served as the foundation of this project, focusing on how integrating these technologies can address systemic vulnerabilities in democratic processes by providing a transparent and resilient architecture for voting.

Electoral Vulnerabilities and Inefficiency in the Philippines

The evolution of electoral systems in the Philippines highlights a persistent struggle between traditional practices and the need for technological integrity. Historically, manual voting was criticized for being labor-intensive, slow, and prone to human error or deliberate manipulation during the canvassing process. While the transition to automated systems like Optical Mark Recognition (OMR) machines improved counting speed, it introduced new vulnerabilities. Reports from the 2022 national elections detailed widespread malfunctions of Vote Counting Machines (VCMs) and faulty SD cards, which fueled public skepticism and allegations of administrative tampering (Tsek.ph, 2022).

Furthermore, the structural reliance on centralized servers creates a "Single Point of Failure" (SPOF). In a centralized architecture, a single successful breach of the main server could allow an adversary to alter the consolidated results of an entire province or nation without leaving an immediate audit trail. This was notably highlighted during the 2016 "Comeleak" incident, where a massive data breach exposed the sensitive information of millions of Filipino voters, proving that centralized repositories are high-value targets for cyber-attacks (Calimbahin, 2011; Philstar, 2016). These systemic weaknesses demonstrate that current automation only speeds up the process but does not necessarily secure the underlying data from sophisticated internal or external threats.

Blockchain Fundamentals and Data Security

Blockchain technology is an **emerging technology** that offers a robust solution to these security gaps through its inherent nature of decentralization and immutability. Grounded in the foundational concept of a peer-to-peer (P2P) network, blockchain eliminates the need for a central intermediary by distributing the ledger across all participating nodes (Nakamoto, 2008). In this framework, data is not stored in one place; rather, every participating computer holds a complete, identical copy of the transaction history.

Each transaction or vote is secured using cryptographic hashing—specifically the **SHA-256 algorithm**. This algorithm converts data into a unique, fixed-length string of characters; even a minor change to the original data (such as changing a single vote) would result in a completely different hash. Because each block contains the "parent hash" of the previous one, the blocks are mathematically chained together. To alter a past vote, an attacker would have to recalculate the hashes for every subsequent block and gain control over at least 51% of the network nodes to validate the fraudulent chain, a feat that is computationally and logistically near-impossible in a properly distributed network (NIST, 2018; Zheng et al., 2017).

Multi-Factor Authentication and Biometric Integration

To complement digital security, embedded systems leverage physical authentication layers to ensure the "one person, one vote" mandate. The integration of **biometric sensors**, such as the AS608 optical fingerprint module, provides a unique physiological identifier that is far more difficult to forge than traditional paper IDs or signatures (Eligo, 2024). Fingerprint biometrics serve as a "Living ID," ensuring that the person present at the terminal is indeed the registered voter.

When combined with Radio Frequency Identification (RFID) technology—where a voter must also tap a registered national ID—the system creates a **Multi-Factor Authentication (MFA)** environment. This hardware-software synergy ensures that the digital ballot is only unlocked after a physical "handshake" between the voter's biometric data and the terminal's local secure database (Al-Ameen & Al-Shaikh, 2021). By processing this verification at the "edge" of the network (on the physical voting terminal itself), the system prevents remote identity theft and ensures that the gateway to the blockchain is strictly guarded by physical presence.

Potential of Blockchain Application to Voting Processes

The application of blockchain to voting processes provides **End-to-End (E2E) Verifiability**, a critical feature for restoring public confidence. Research indicates that blockchain-based e-voting systems can effectively manage large-scale elections by utilizing **Smart Contracts**—self-executing code that automatically tallies votes as they are cast, thereby reducing human intervention and the risk of "mathematical errors" during canvassing (Hjálmarsson et al., 2018; Kshetri & Voas, 2018).

While individual voter anonymity is preserved through advanced encryption, the encrypted transaction hashes are accessible on a public ledger. This allows observers to verify in real-time that the number of votes cast matches the number of voters who checked in, without seeing who each person voted for. Furthermore, the system generates a digital receipt—a unique transaction hash—that the voter can use to cross-reference their entry against the public record. This "receipt" does not show the candidate's name (to prevent vote-selling), but it proves that the vote was successfully "anchored" into the blockchain and counted in the final results, ensuring absolute transparency in the electoral outcome (World Economic Forum, 2020).

HARDWARE DESIGN

The hardware architecture of VOTECHAIN is designed to prioritize security, redundancy, and user accessibility. By integrating distinct modules for input, processing, and output, the system ensures that voter authentication is robust and that feedback is immediate. The design centers around the Raspberry Pi 5, which acts as the high-performance controller capable of managing biometric data processing and cryptographic operations simultaneously.

List of Components

Sensor and Actuators

- **AS608 Optical Fingerprint Sensor**: Used for biometric enrollment and verification. It connects via UART serial communication to secure voter identity.
- **MFRC522 RFID Reader**: A 13.56MHz contactless reader used to scan voter ID cards. It uses the SPI communication protocol for fast data transfer.
- **Active Buzzer**: Provides audible feedback for successful scans or error alerts.

Other Components

- **Raspberry Pi 5**: The central controller handling the UI, peripheral management, and API communication.
- **5-inch Touchscreen Display**: The primary visual interface for the voting kiosk. It allows voters to view the ballot, navigate options, and receive visual feedback during the voting process.
- **TM1637 4-Digit 7-Segment Display:** An external counter that displays the real-time vote count, providing immediate public transparency independent of the main screen.

- **4x3 Membrane Matrix Keypad**: Allows users to input their unique ID numbers during registration or as a backup input method.
- **Power Supply (USB-C 27W)**: Ensures stable power delivery to the Pi and connected peripherals.
- **Breadboard and Jumper Wires**: Used for connecting the sensors/actuators and other components together.

Block Diagram

The system follows a simple three-step process to ensure security. First, the Voting Terminal scans the voter's ID and fingerprint, sending this data to the central server to confirm they are eligible to vote. Once the user selects a candidate, the Central Server acts as the secure controller; it immediately locks the voter's profile to prevent double-voting and saves the vote into the blockchain. Instead of just saving the number, the system generates a unique 'digital fingerprint' (hash) for that vote and links it to the previous one, creating an unbreakable chain. Finally, the Public Dashboard receives these updates instantly via the network, allowing anyone to see the vote count change in real-time and verify the election data

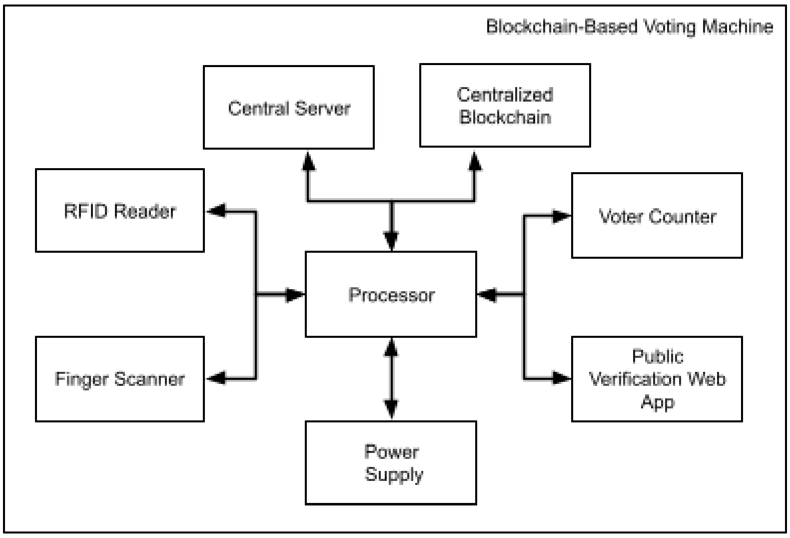


Figure 1. Block Diagram for Blockchain-Based Voting Machine

Circuit Diagram

To ensure signal integrity and prevent bus contention, the design leverages the Raspberry Pi 5's dedicated hardware controllers rather than software-simulated protocols.

1. **Biometric Interface (UART - Universal Asynchronous Receiver-Transmitter)** - The AS608 Fingerprint Sensor requires a reliable, bi-directional data stream for sending template packets. We utilize the Pi's hardware UART to provide a dedicated TX/RX channel, eliminating the risk of collision with other peripherals.
2. **RFID Interface (SPI - Serial Peripheral Interface)** - The MFRC522 RFID Module handles high-speed tag reading which requires synchronous clock timing. The SPI bus is selected for its high throughput, using a dedicated Chip Select (CE0) line to prevent bus contention while allowing rapid data transfer necessary for a seamless "tap-and-go" experience.
3. **Manual Input (Matrix GPIO)** - To conserve GPIO pins, the 4x3 Keypad utilizes a row-column scanning matrix. This method allows 12 distinct inputs to be read using only 7 pins by rapidly strobing the row lines and reading the column states.
4. **Feedback & Display (Direct GPIO)** - Output devices are driven by direct digital signaling. The TM1637 Display utilizes a custom 2-wire protocol (Clock + Data) separate from the high-speed SPI/UART buses to prevent timing interference.

The figure and table below shows the complete circuit design of the project alongside its pin diagrams.

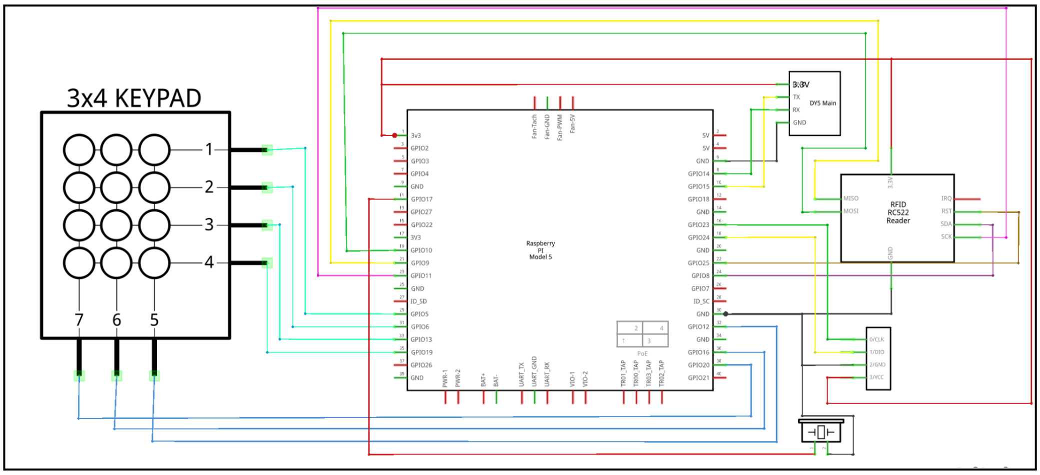


Figure 2. Circuit Diagram of VOTECHAIN

Table 1. VOTECHAIN Pin Diagram Table

Component	Pin Name	RPi GPIO (BCM)	Physical Pin (Header)
RFID (RC522)	SDA (SS)	GPIO 8 (CE0)	Pin 24
	SCK	GPIO 11 (SCLK)	Pin 23
	MOSI	GPIO 10	Pin 19
	MISO	GPIO 9	Pin 21
	RST	GPIO 25	Pin 22
	GND	Ground	Pin 6
	3.3V	3.3V	Pin 1
Fingerprint (AS608)	RX (Green)	GPIO 14 (TXD)	Pin 8
	TX (White)	GPIO 15 (RXD)	Pin 10
	VCC	3.3V	Pin 1
	GND	Ground	Pin 6
Matrix Keypad	Row 1	GPIO 5	Pin 29
	Row 2	GPIO 6	Pin 31
	Row 3	GPIO 13	Pin 33
	Row 4	GPIO 19	Pin 35

	Col 1	GPIO 12	Pin 32
	Col 2	GPIO 16	Pin 36
	Col 3	GPIO 20	Pin 38
TM1637 Display	CLK	GPIO 23	Pin 16
	DIO	GPIO 24	Pin 18
	VCC	3.3V	Pin 1
	GND	Ground	Pin 6
Active Buzzer	Signal (+)	GPIO 17	Pin 11
	GND (-)	Ground	Pin 6

Prototype Layout

For this specific prototype implementation, a custom Printed Circuit Board (PCB) was not utilized. Instead, the hardware was assembled using a solderless breadboard to maximize flexibility during the testing and debugging phases.

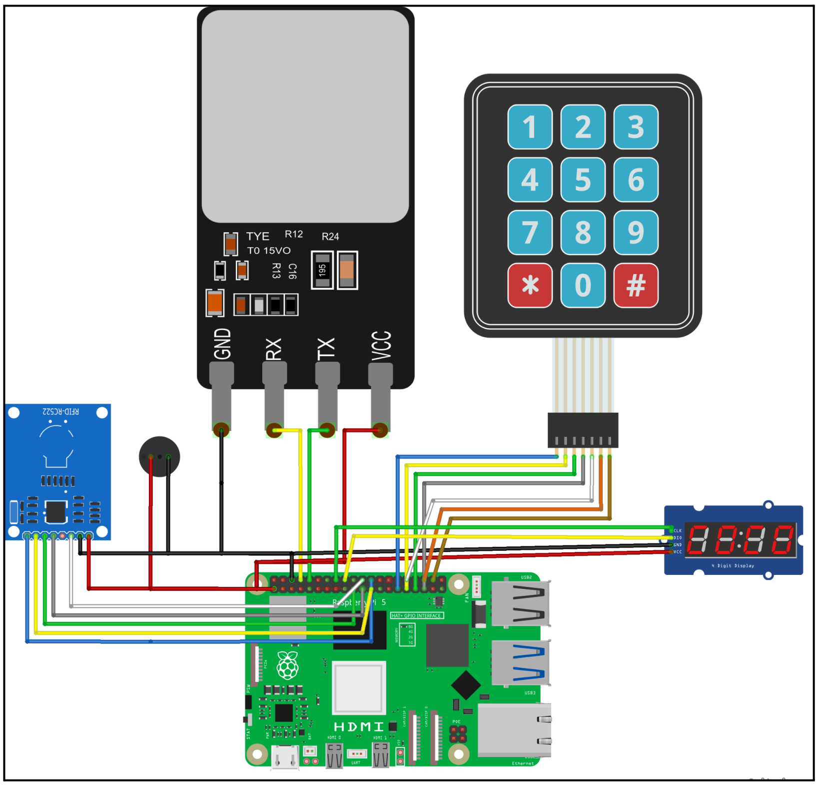


Figure 3. Wiring Diagram of VOTECHAIN

To transition the prototype from a bench-top circuit to a functional voting kiosk, the components were housed within a custom 3D-printed enclosure. The chassis was fabricated using Polylactic Acid (PLA) filament via Fused Deposition Modeling (FDM) 3D printing, selected for its rigidity and ease of manufacturing.

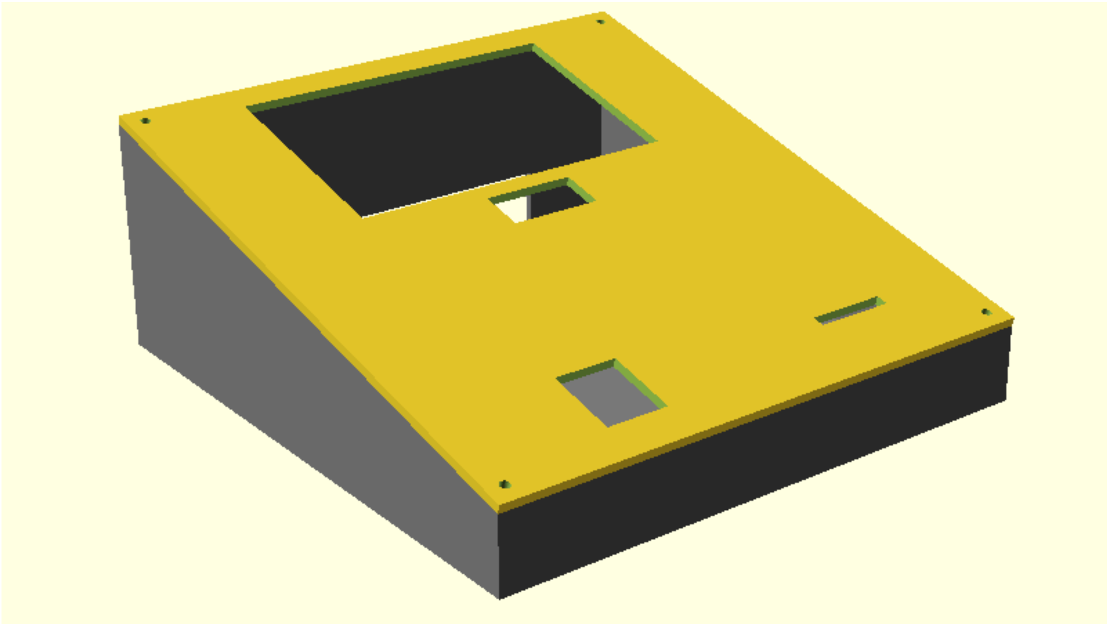


Figure 4. VOTECHAIN Hardware Enclosure

Figure 5. Assembled Hardware Enclosure with Full Components

USER INTERFACE

The VOTECHAIN project is divided into three parts: a Voting Terminal for registering voters and casting votes; a publicly available and auditable ledger, and the actual Blockchain-Based Voting system. This section focuses on the first two components, the frontend interfaces through which users interact with the system. The design prioritizes security, accessibility, and transparency, ensuring that voters can cast their ballots with confidence while observers can audit the results in real-time.

User Interface Flow

The interaction flow is designed to be linear and restrictive for the voter to prevent errors, while being open and exploratory for the public observer to encourage auditing.

Voting Kiosk

The Voting Kiosk serves as the primary entry point for casting votes. Powered by a Raspberry Pi 5 running a custom Python CustomTkinter application, the interface guides the voter through a strict six-stage process. This flow is enforced by the software to ensure that each step (authentication, selection, and submission) is completed successfully before the next step is unlocked.

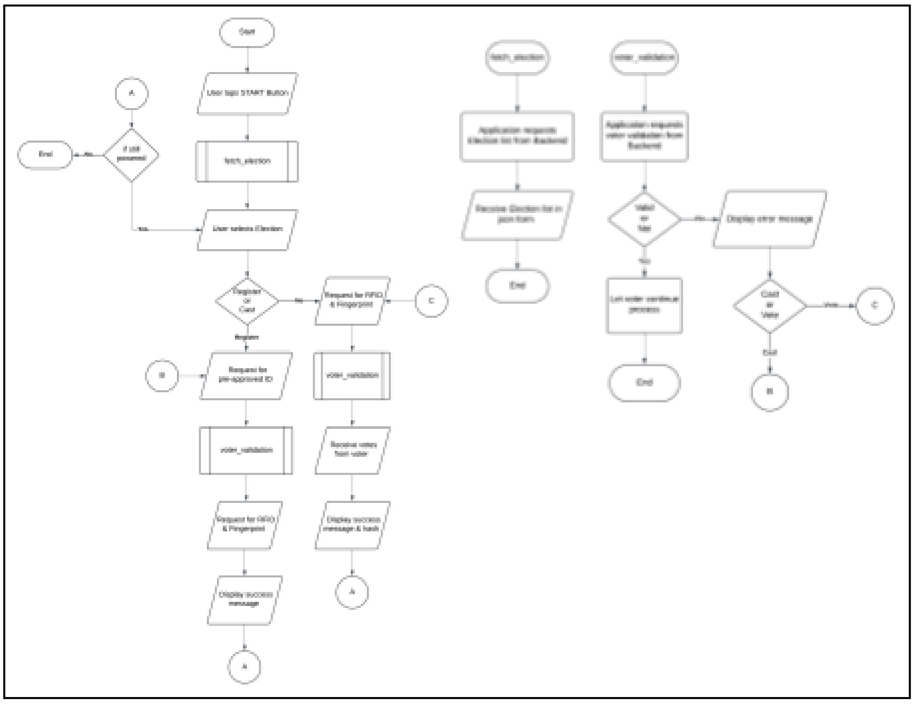


Figure 6. Voting Kiosk User Flow

1. The system begins in a secure idle state, displaying the VOTECHAIN logo and a "TAP TO START" button.
2. The terminal retrieves active elections. Active elections are highlighted, while closed elections are dimmed.
3. Multi-Factor Authentication:
    a. Step 1 (RFID): User taps their ID card. The system hashes the UID.
    b. Step 2 (Biometrics): User scans their fingerprint. The system verifies the template.
    c. Both credentials are sent to the server to check registration status.
4. The ballot renders dynamically. Single-choice positions use radio buttons, while multi-choice positions use checkboxes with limit enforcement.
5. The voter submits the ballot. The data is encrypted and sent to the blockchain.

6. A confirmation screen displays the unique Blockchain Transaction Hash, allowing the voter to verify their vote later.

Public Verification Web App

The Web Application serves as the transparency layer for VOTECHAIN. It is designed to allow any citizen to audit election activities in real-time without the need for an account or any verification. The observer workflow consists of three primary interactive modules:

1. **Real-Time Election Dashboard:** The election dashboard which upon accessing the platform, users are presented with a high-level status board for all elections.
    a. The interface displays an "Active/Live" badge to indicate ongoing network activity.
    b. The dashboard maintains a persistent connection to the backend. This allows key metrics such as Total Votes Cast and Voter Turnout Percentage to update automatically on the client side as soon as a new vote is successfully received.
2. **Immutable Public Ledger**: A dedicated "Public Ledger" panel provides a window into the underlying blockchain structure. This feature is critical for the system's auditability.
    a. The ledger lists all blocks in reverse chronological order. Users can click to expand any individual block to view its contents.
    b. Within the expanded block view, the system displays the decrypted vote contents (e.g., "President: Candidate A") alongside the timestamp. This proves that the vote was counted correctly while preserving voter anonymity by not displaying any personal identifiers.

c. The full SHA-256 hash is visible, allowing voters to cross-reference their personal digital receipts against the public record.

3. **Results Visualization**: Voting results are displayed to provide an immediate overview of the election status.
   a. As new votes are ingested from the blockchain, the results display updates immediately to reflect the new standings.
   b. Each candidate is listed with their party affiliation and current vote count, allowing observers to track the progress of the election at a glance.

UI/UX Design

The User Interface (UI) and User Experience (UX) design strategy for VOTECHAIN centers on "Security through Clarity." Given the critical nature of elections, the interface must not only look modern but also instill immediate trust. The design language unifies the embedded hardware and the web platform under a cohesive "Cyber-Security" aesthetic, utilizing high-contrast dark modes to reduce eye strain in low-light voting booths and minimize power consumption.

A unified color palette is strictly enforced across both the Python-based Kiosk and the React-based Web App to maintain consistent branding. Each color plays a functional role in guiding user behavior:

- **Deep Navy Background (#050511)**: Unlike pure black, this deep blue hue reduces halation (text glare) on screens while providing a stark canvas that makes colored elements pop. It establishes a serious, professional tone associated with secure environments.
- **Electric Cyan Primary (#00E5FF)**: This high-energy color is reserved strictly for Primary Actions (e.g., "Cast Vote," "Next") and Live Indicators. Its

luminosity draws the eye instantly, guiding the user through the happy path of the application.

- **Cyber Purple Secondary (#9D4EDD)**: Used for structural elements, borders, and static headers. It separates content sections without competing for attention against the primary actions.
- Functional Status Colors:
  - **Signal Green (#00C853)**: Reserved for "Success" states (e.g., "Fingerprint Verified"). It provides immediate psychological reassurance.
  - **System Red (#FF4444)**: Used for critical errors or blocking actions.
  - **Alert Orange (#FF9E00)**: Indicates state changes like "Election Ended", ensuring users distinguish status updates from system failures.

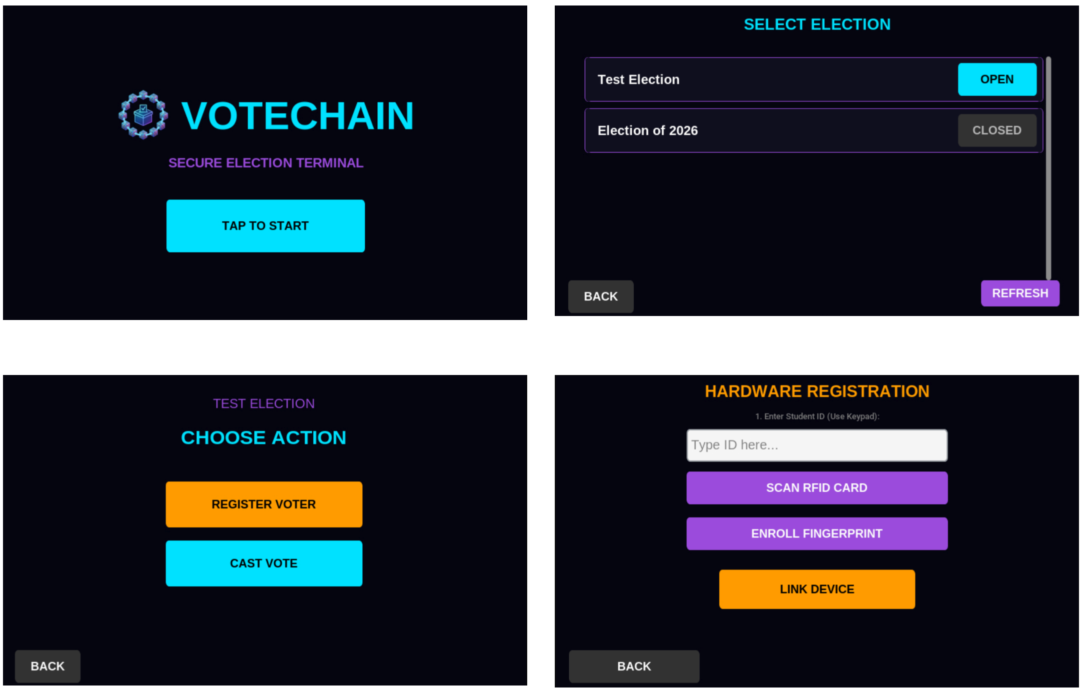
The Kiosk interface, built with CustomTkinter, is optimized specifically for the 5-inch touchscreen of the Raspberry Pi. The UX prioritizes accessibility and error prevention:

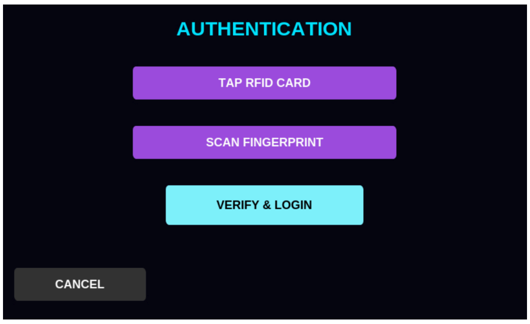
- Standard buttons are oversized (typically 300px width by 80px height) with ample padding. This accommodates users with varying degrees of motor control and prevents accidental mis-taps on the small screen.
- A key challenge in embedded systems is the "invisible" nature of hardware inputs (e.g., waiting for an RFID tap). The UI compensates for this by dynamically updating button states. When the hardware enters scanning mode, buttons shift to an "Active/Listening" state (e.g., changing text from "SCAN" to "WAITING..."). Upon a successful hardware interrupt, the UI instantly transitions to Green, providing visual confirmation that the physical action was registered (See Testing and Evaluation Section).
- Instead of subtle toast notifications which might be missed, errors (such as "Fingerprint Mismatch") trigger high-contrast Modal Popups. These block the

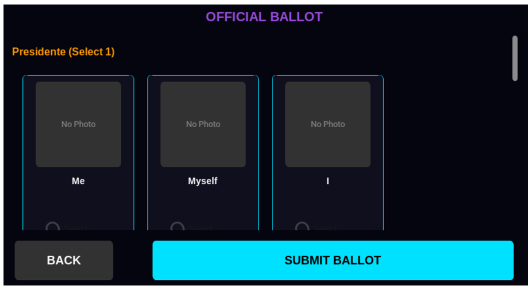
rest of the interface, forcing the user to acknowledge the issue before proceeding, thus preventing confusion (See Testing and Evaluation Section).

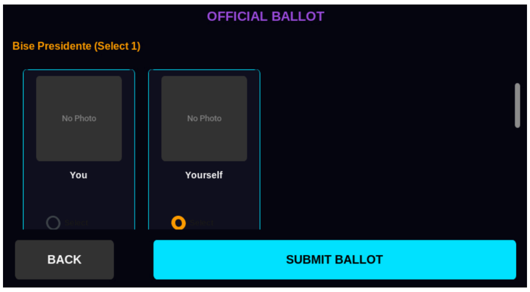
The table below shows the interaction that the user will perform with the kiosk. It begins with the Start page, which initializes the session upon a "Tap to Start" interaction. This transitions to the Election Selection page, where active elections are highlighted in Cyan for visibility, while inactive ones are dimmed, guiding the user to valid choices. Upon selection, the user navigates through the Action Menu, a secure routing hub that allows for either voter registration or ballot casting. For administrators, the Registration interface provides tools to link hardware tokens to student IDs, capturing RFID UIDs and biometric templates. For general voters, the flow moves to the Authentication gatekeeper, which enforces a strict multi-factor check requiring an RFID tap followed by a fingerprint scan. Once authenticated, the ballot interface is dynamically rendered based on the election's configuration, employing radio buttons for single-choice positions and checkboxes for multi-choice roles.

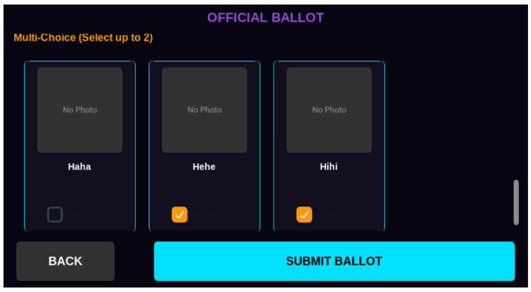
Table 2. Voting Kiosk User Interface











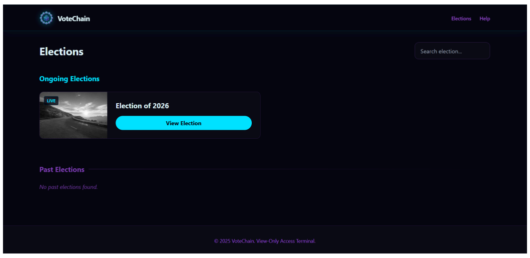
Meanwhile, the web app was developed using React and Tailwind CSS. It focuses on presenting the blockchain data in an accessible, digestible format.The user experience begins at the Election List, which acts as the platform's landing page. Here, active elections are distinguished by vibrant "LIVE" badges and full-color thumbnails, while past elections are visually deprioritized using grayscale filters to reduce cognitive load.

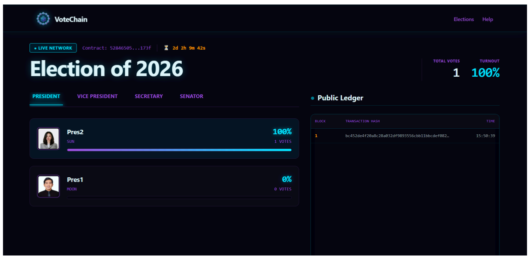
Upon entering a specific election dashboard, the user is immersed in a real-time data environment. The layout is divided into two primary zones: the Live Results column and the public ledger panel. The live Results section uses dynamic "race bars" to visualize candidate standings; these bars animate smoothly as new votes are linked, providing immediate visual feedback on the election's progress. Instead of requiring manual page refreshes, the interface utilizes WebSockets to push updates instantly, reinforcing the "live" nature of the event.

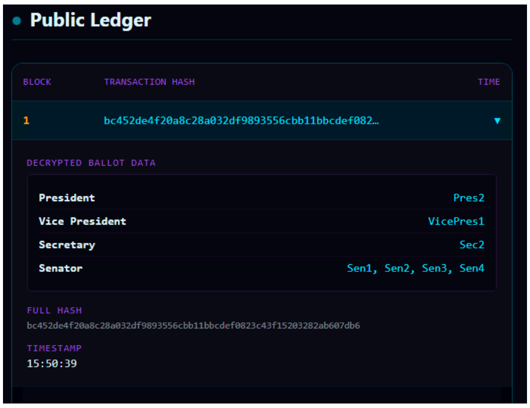
The public ledger serves as the core transparency mechanism. Designed to look like a scrolling terminal log, it lists every block in reverse chronological order.

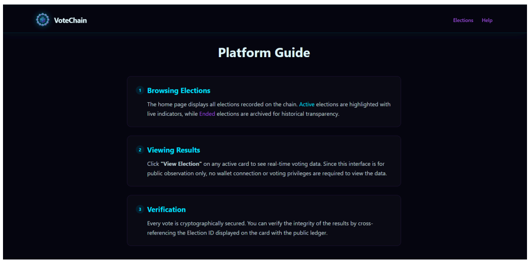
Interactive elements allow users to expand individual blocks to reveal decrypted ballot data and timestamps, proving that votes are being counted correctly.

Table 3. Public Web App User Interface









RESULTS AND DISCUSSION

Project Features and Functions

The VOTECHAIN prototype successfully implemented the proposed dual-interface architecture, integrating secure hardware inputs with a transparent blockchain backend. The following core features were developed and tested:

1. **Secure Voter Kiosk** - The system effectively demonstrated Multi-Factor Authentication (MFA) to enforce the "One Person, One Vote" policy.
    a. **RFID Integration**: The MFRC522 reader successfully captured unique ID card tokens, which were immediately hashed (SHA-256) to prevent storing raw ID data.
    b. **Biometric Verification**: The AS608 optical sensor successfully enrolled new fingerprints and verified returning voters. The system's logic correctly prevented users from voting twice by cross-referencing their encrypted credentials against the backend database before unlocking the ballot.
    c. **Feedback Mechanisms:** The 5-inch touchscreen and active buzzer provided clear visual and auditory feedback during the scan process, reducing user error rates.
2. **Digital Ballot Interface** - The CustomTkinter-based kiosk application demonstrated a responsive and adaptable ballot system.
    a. **Dynamic Rendering**: The application successfully fetched election configurations from the server and rendered the appropriate UI elements (Radio Buttons for single-choice, Checkboxes for multi-choice) in real-time.

   b. **Vote Limitation Logic**: The interface successfully prevented "over-voting" (e.g., selecting more senators than allowed) by showing an error message if the user tries to submit their ballot.

3. **Immutable Blockchain Storage** - The custom Python blockchain implementation achieved its primary goal of tamper-evident storage.
   a. **Proof of Authority**: Upon vote submission, the system executed a secure hashing algorithm (SHA-256) to generate a block hash that mathematically links to the previous entry, establishing a continuous chain of custody.
   b. **Transaction Hashing**: Every vote generated a unique transaction hash, which was displayed to the voter as a digital receipt. Attempts to manually alter the JSON ledger file resulted in chain validation errors, proving the system's immutability.

4. **Real-Time Public Transparency (Web App)** - The React-based web dashboard provided a live window into the election process without compromising voter privacy.
   a. **WebSocket Updates**: The system achieved sub-second latency between a vote being cast on the Pi and the "Total Votes" counter updating on the web dashboard.
   b. **Public Ledger**: The "Public Ledger" feature successfully displayed decrypted ballot data (e.g., "President: Candidate A") while keeping the voter's identity anonymous, satisfying the requirement for public auditability.
   c. **Hardware Counter**: The integration of the TM1637 7-segment display provided an additional layer of physical transparency, allowing observers in the room to see the vote count increment instantly, independent of the digital screens.

Testing and Evaluation

To validate the reliability and robustness of the VOTECHAIN system, a series of unit tests and integration tests were conducted. These tests focused on error handling, security enforcement, and system performance under various conditions.

The first test case focused on the resilience of the public web application when accessing invalid data. When a user attempts to navigate to a URL for a non-existent election (e.g., /election/9999). The system should intercept the 404 error and display a custom "Block Not Found" page instead of crashing. Based on the figure below, this test case passed successfully.

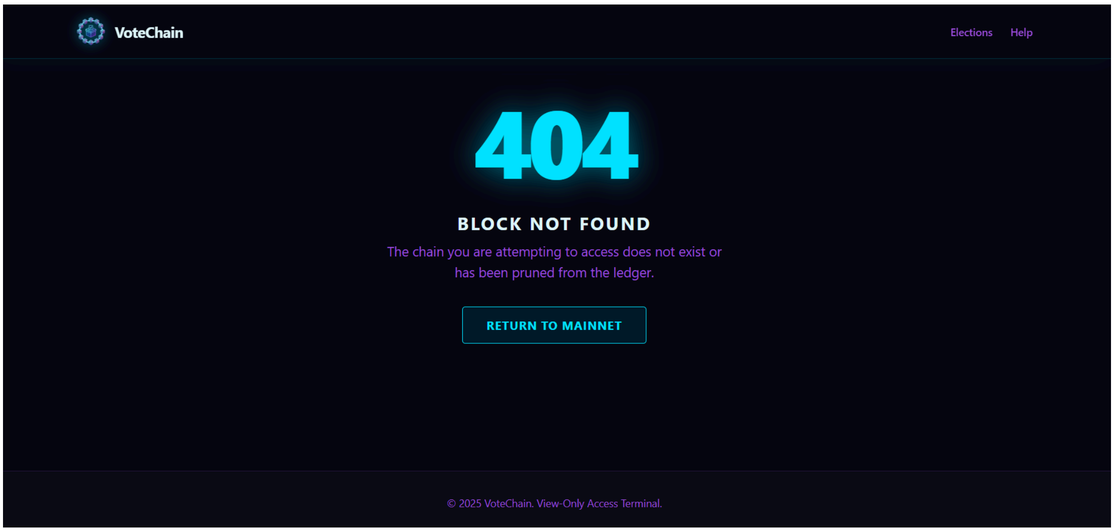


Figure 7. Web App Invalid Election ID Page

The registration process was tested against multiple failure conditions to ensure that only valid, unique voters could be enrolled in the system.

Table 4. Voter Registration Test Cases

Scenario	Procedure	Expected Result	Actual Result	Status
Empty Fields	Admin attempts to click "Link Device" without entering a Student ID or	System prevents submission and shows a popup error: "Please	Error popup displayed; API request was not sent.	**PASSED**

	scanning any hardware.	complete all fields".		
ID Not Authorized	Admin enters a Student ID that does not exist in the university database.	System returns an error: "Student ID not found or not eligible".	The backend returned 404; Kiosk displayed "ID Invalid".	**PASSED**
ID Already Registered	Admin enters a valid Student ID that has already been linked to a device.	The system rejects the registration to prevent duplicate accounts.	Error popup displayed: "User already registered".	**PASSED**
RFID Card Conflict	Admin scans an RFID card that is already linked to another user.	System detects the duplicate UID hash.	Error popup displayed: "RFID Card already in use".	**PASSED**
Fingerprint Conflict	Admin scans a fingerprint that is already enrolled in the AS608 memory.	The sensor's internal logic or backend check detects the duplicate template.	Error popup displayed: "Duplicate Fingerprint Detected".	**PASSED**
Successful Registration	Admin enters a valid new ID, scans a new RFID card, and enrolls a new fingerprint.	System displays "Success", links the hardware tokens, and saves the user to the database.	"Hardware Linked Successfully" message displayed.	**PASSED**

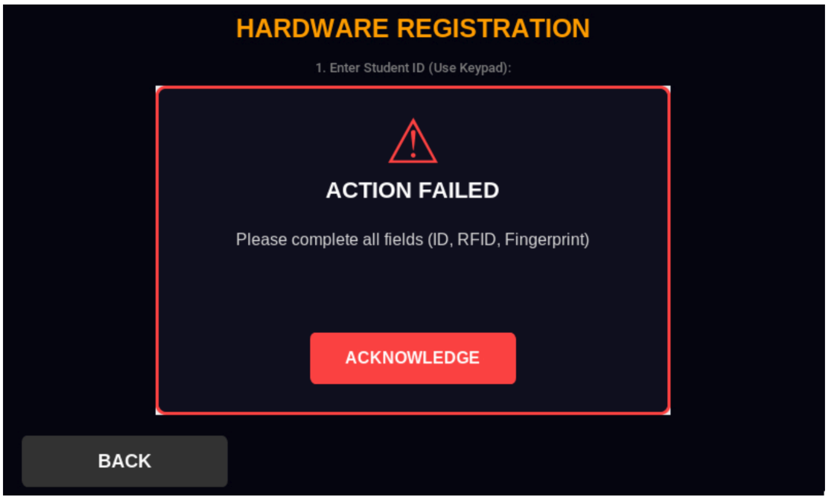


Figure 8. Empty Fields Scenario

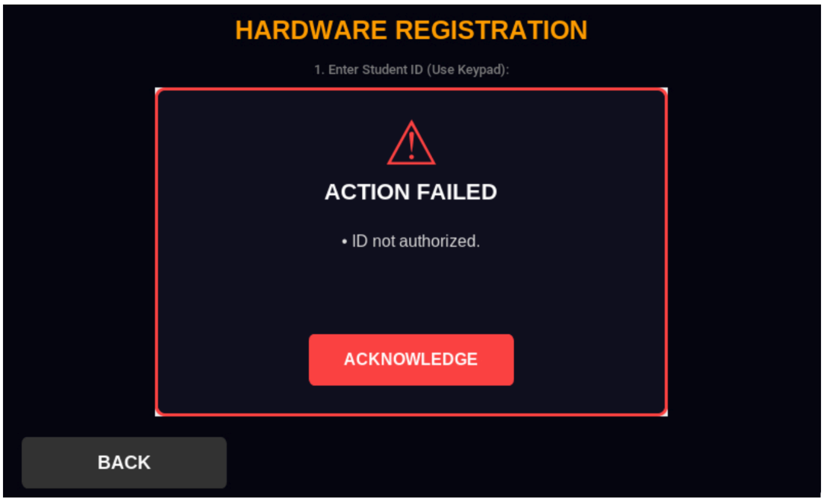


Figure 9. ID Not Authorized Scenario

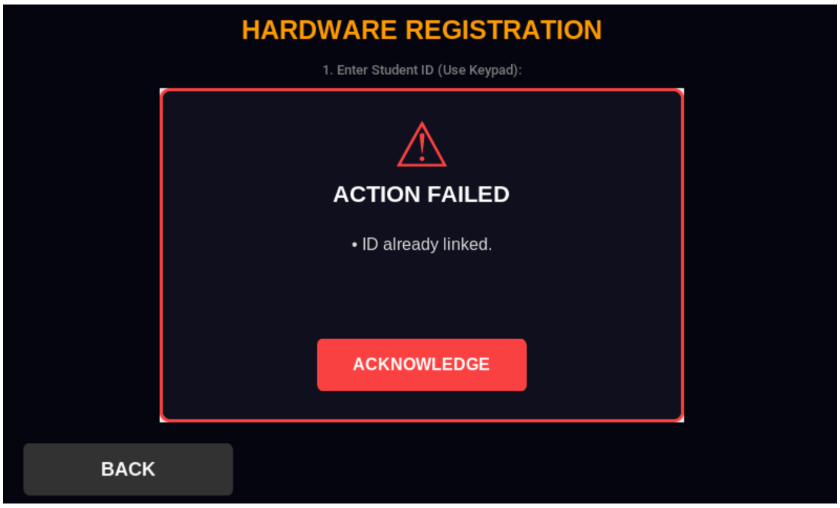


Figure 10. ID Already Registered Scenario

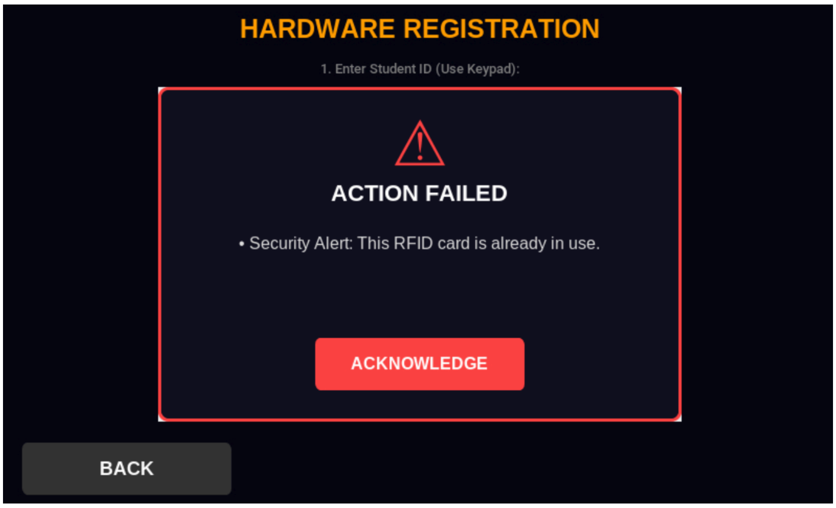


Figure 11. RFID Card Conflict Scenario

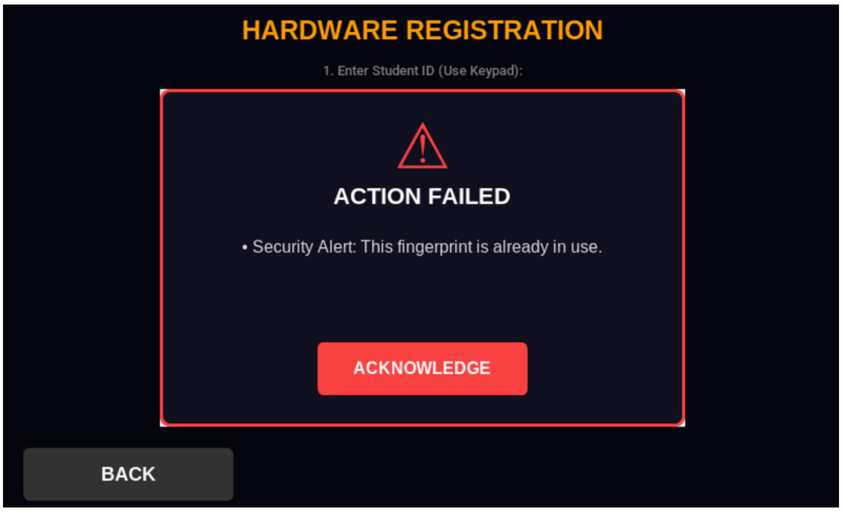


Figure 12. Fingerprint Conflict Scenario

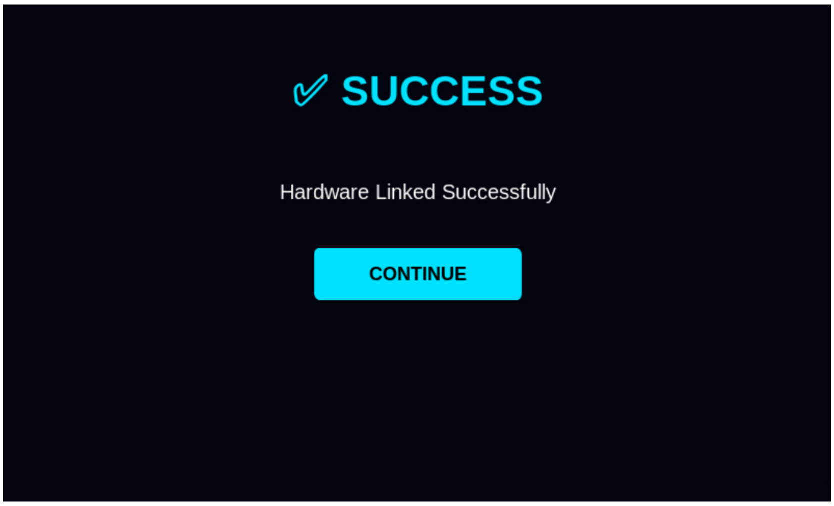


Figure 13. Successful Registration Scenario

The voting interface was tested to ensure the integrity of the "One Person, One Vote" principle and to validate ballot inputs.

Table 5. Vote Casting Test Cases

Scenario	Procedure	Expected Result	Actual Result	Status
Incomplete Auth	User attempts to click "Verify & Login" without completing both RFID and Fingerprint scans.	The system prevents login and prompts user to complete scans.	Error popup: "Please complete both scans first".	**PASSED**
Duplicate Voter	A user who has already voted attempts to authenticate again.	Backend checks the blockchain/database and denies access.	Error popup: "Voter has already cast a ballot".	**PASSED**
Unregistered Voter	A user with an unknown RFID card or fingerprint attempts to log in.	The system denies access.	Error popup: "Credentials not found".	**PASSED**
Empty Ballot	User attempts to submit a ballot without selecting any candidate.	The system prevents submission (if configured to require votes) or submits a blank vote (abstain).	The system prevented submission until at least one selection was made.	**PASSED**
Over-Voting	User attempts to select more options than allowed (e.g., 5 Senators for a 4-seat position).	The system prevents the ballot from being submitted.	Error popup: “Too many selections for <position_name>.	**PASSED**

Successful Vote	Valid user authenticates, fills the ballot correctly, and submits.	The vote is encrypted, linked, and a receipt is shown.	"Vote Secured" screen displayed with valid transaction hash.	**PASSED**

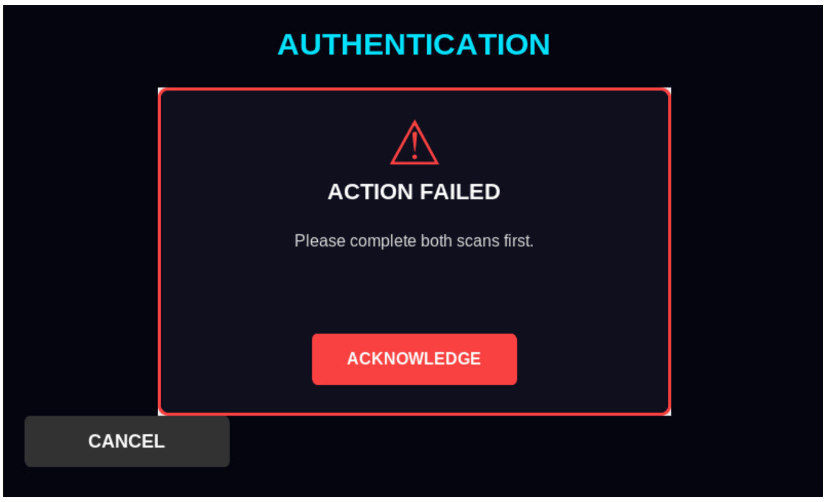


Figure 14. Incomplete Authentication Scenario

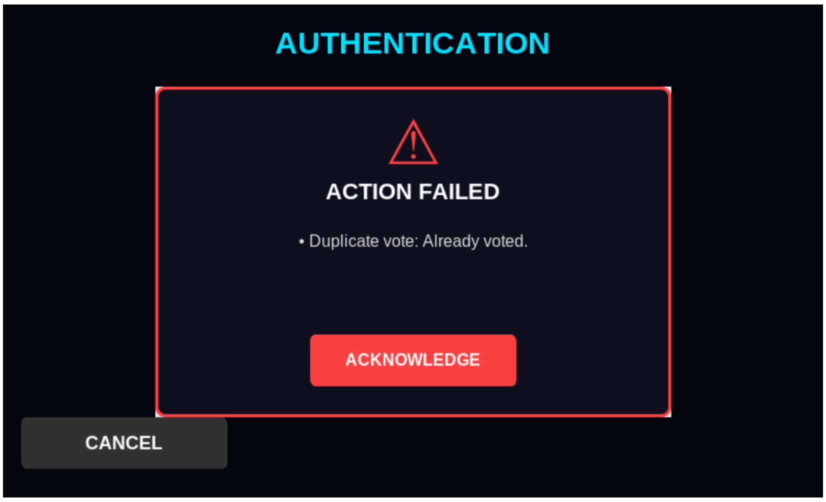


Figure 15. Duplicate Voter Scenario

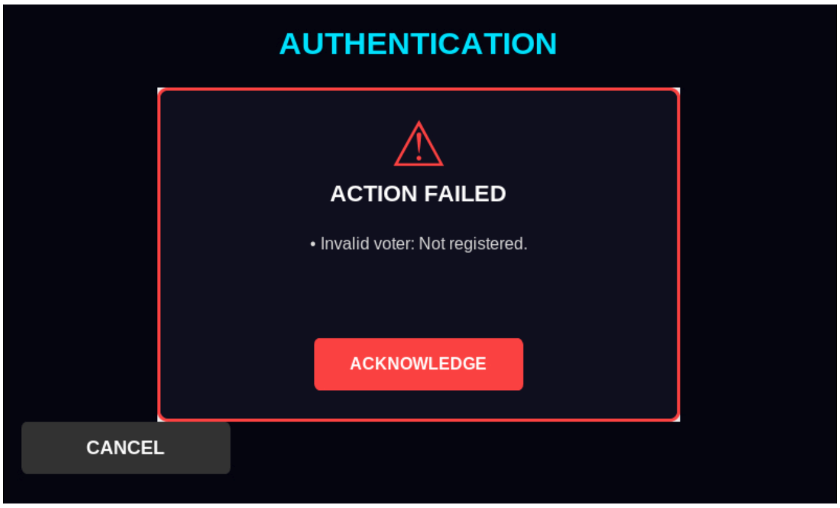


Figure 16. Unregistered Voter Scenario

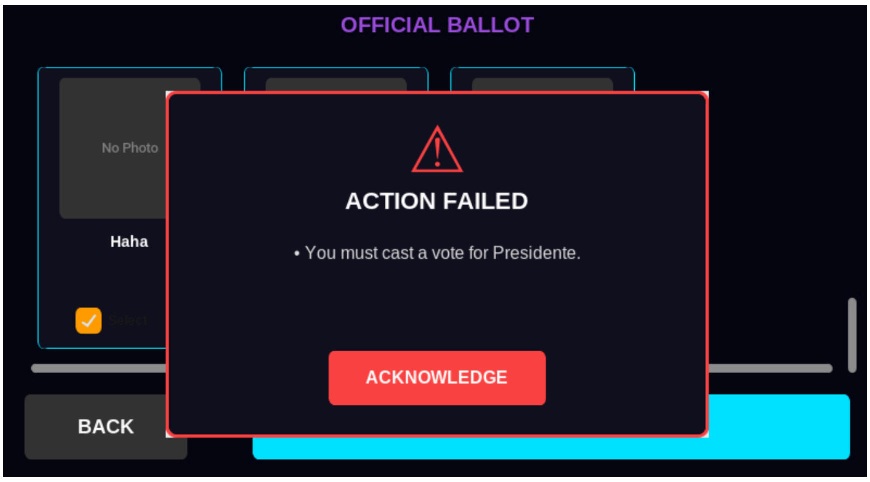


Figure 17. Empty Ballot Scenario

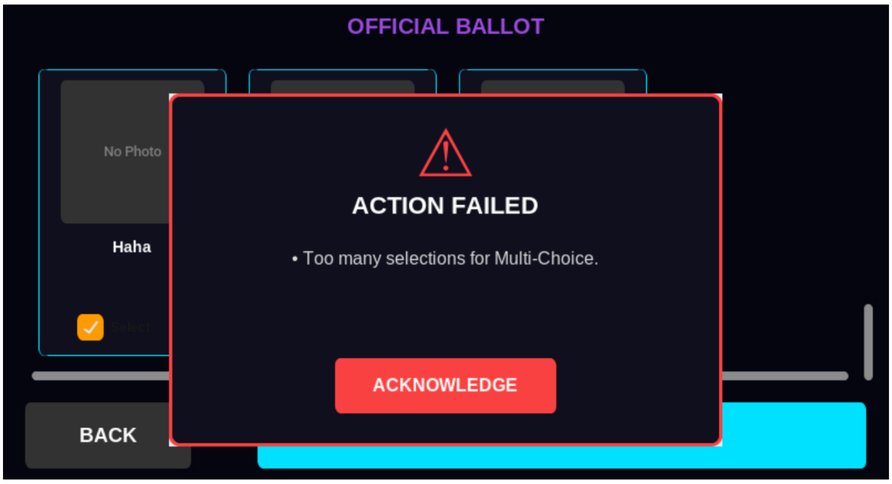


Figure 18. Over-Voting Scenario

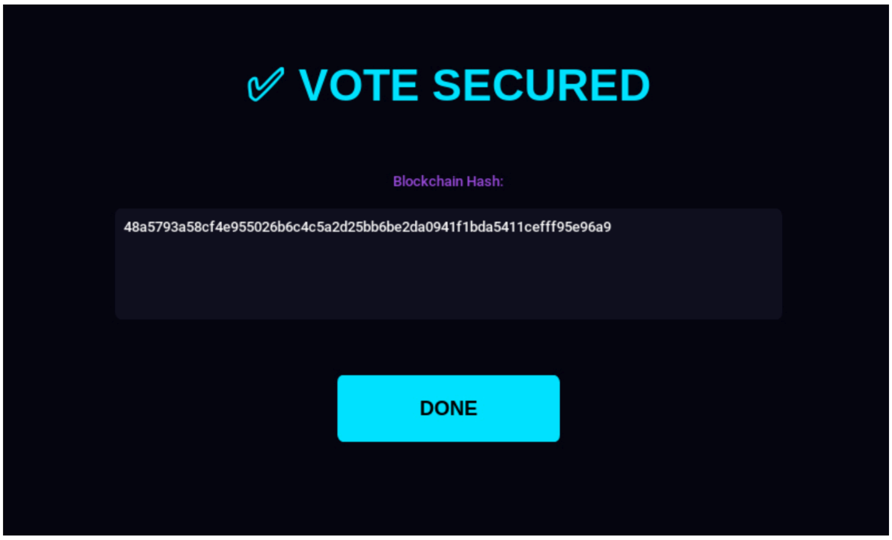


Figure 19. Successful Vote Scenario

To assess the practicality of the system for real-world deployment, the students measured key performance indicators (KPIs) during a mock election.

Table 6. Performance Metrics during Mock Election

Metric	Measured Value	Description
Average Voting Duration	< 1 minute (Depends on Ballot length)	Measured from the moment the user taps "Start" to the display of the "Vote Secured" receipt.
Biometric False Rejection Rate (FRR)	~30%	Approximately 3 in 10 fingerprint scans required more than one attempt due to finger placement issues.
Block Finalization Latency	1.2 seconds	The average time taken for the system to execute the cryptographic hashing algorithm, link the new block to the chain, and permanently commit the transaction to the immutable ledger.
Web Socket Latency	< 500ms	The delay between the block being linked on the Kiosk and the "Total Votes" counter updating

		on the public web dashboard.

Summary of Peer Evaluation

To assess the functionality, usability, and reliability of the VOTECHAIN system, a peer evaluation was conducted on January 14, 2025. The study involved thirty-five (35) College of Engineering and Information Technology (CEIT) students from Cavite State University - Main Campus who served as respondents. Each evaluator was provided with a "Peer Review - Prototype Evaluation Form" and was tasked with interacting with the voting kiosk and the public dashboard to rate the system’s performance and provide qualitative feedback.

To quantify the project's success, evaluators rated the system across four key performance indicators using a scale of 1 to 10, where 10 represents excellent performance. The data collected indicates a high level of approval for the system's functionality and presentation.

Table 7. Summary of Peer Evaluation Ratings (n=35)

Criteria	Average Rating	Description of Metric
Explanation	**9.77 / 10**	Measures how clearly the group articulated the innovation’s working principles and technical mechanisms.
Demonstration	**9.82 / 10**	Assesses the execution of the live demo, including the smoothness of the voting process.
Quality	**9.63 / 10**	Evaluates the overall build quality, aesthetic design, and craftsmanship of the hardware prototype.

Effectiveness	**9.77 / 10**	Rates the system's ability to solve the identified problem of electoral fraud and inefficiency.
TOTAL AVERAGE	**9.75 / 10**	Overall Performance

The VOTECHAIN prototype achieved an outstanding **Total Average of 9.75 / 10** across all core evaluation categories, indicating a high level of technical and operational excellence. This performance is anchored by the **Demonstration (9.82/10)** and **Effectiveness (9.77/10)** metrics, which confirm that the system is not only operationally seamless during live execution but also highly relevant in addressing electoral fraud. Furthermore, the high rating in **Explanation (9.77/10)** validates the group's ability to clearly articulate complex blockchain principles, while the **Quality (9.63/10)** score confirms that the hardware craftsmanship met the aesthetic and structural expectations of a peer group of engineering students. Collectively, these results demonstrate a successful convergence of technical articulation, physical design, and functional utility.

The specific breakdown and discussion of the evaluation results are shown below.

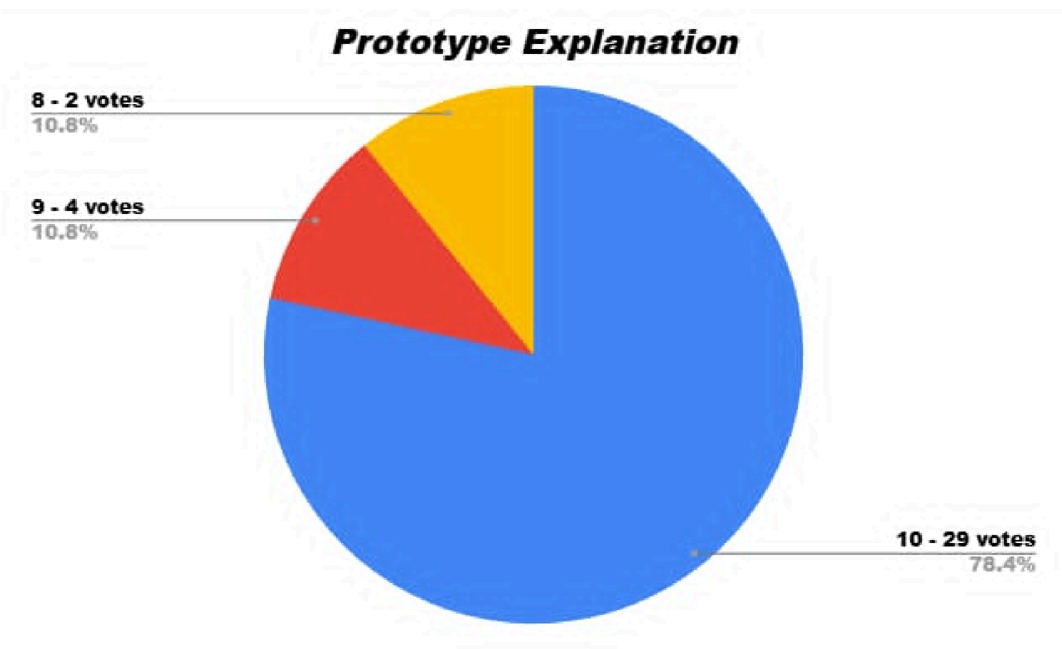


Figure 20. Prototype Explanation

The concentration of scores at the highest level (78.4% at a rating of 10) indicates that the group successfully communicated the complex technical mechanisms of the VOTECHAIN system. The evaluators found the articulation of the working principles involving the blockchain ledger logic and hardware sensor integration to be exceptionally clear. With no score lower than an 8, the data confirms that the technical presentation was both comprehensive and easily understood by a peer group of engineering students.

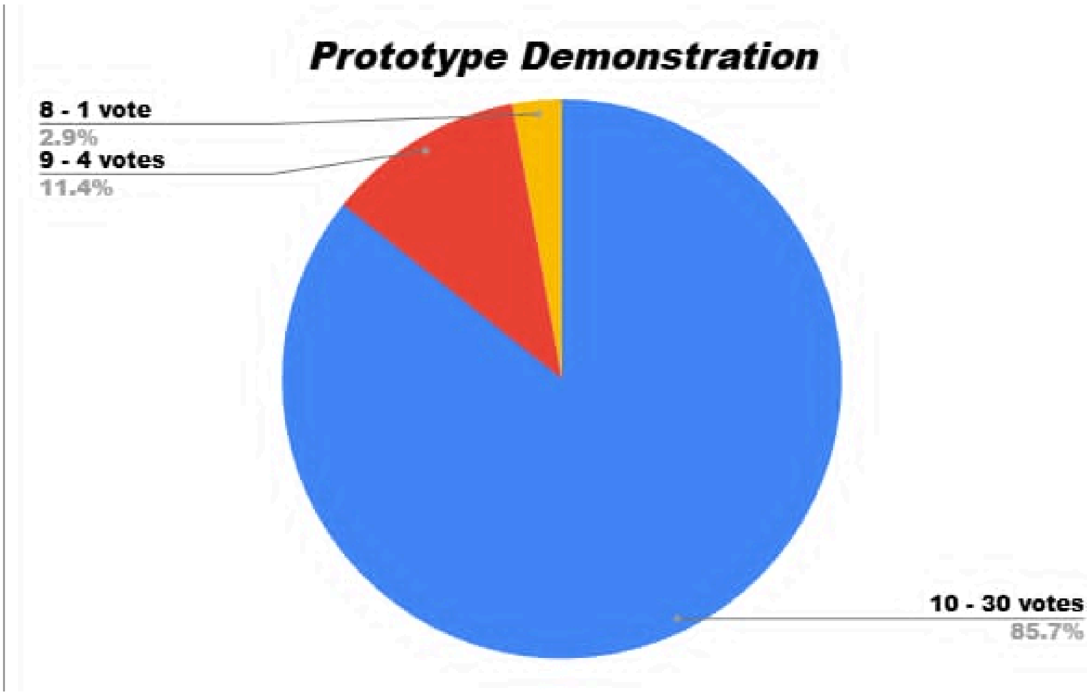


Figure 21. Prototype Demonstration

With 85.7% (30 votes) of the evaluators giving a perfect 10, the demonstration proved that the system is stable and operationally sound. This high rating confirms that the "smoothness" of the voting process—encompassing biometric verification, dashboard accessibility, data transmission to the backend, and the immediate hashing of the vote—was flawlessly executed. The minimal variance in scores highlights the reliability of the system under live conditions especially due to internet conditions, suggesting that the hardware and software integration is seamless enough for practical deployment.

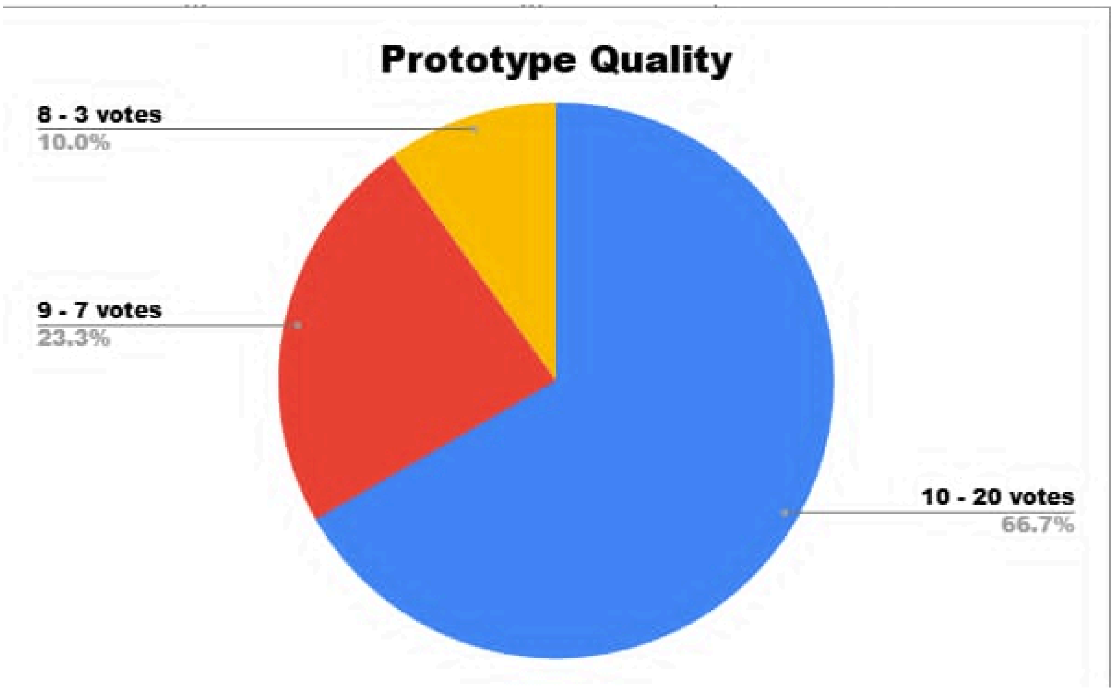


Figure 22. Prototype Quality

The results indicate a high level of appreciation for the physical realization of the VOTECHAIN kiosk. With **66.7%** of evaluators awarding a perfect score for craftsmanship, the data suggests that the prototype's aesthetic design and structural integrity were professional and well-executed. The inclusion of ratings at 9 and 8 reflects a critical peer perspective on the "fit and finish" of the hardware, yet the overall weighted average remains exceptionally high, confirming that the physical prototype successfully mirrors the high-tech nature of the underlying software.

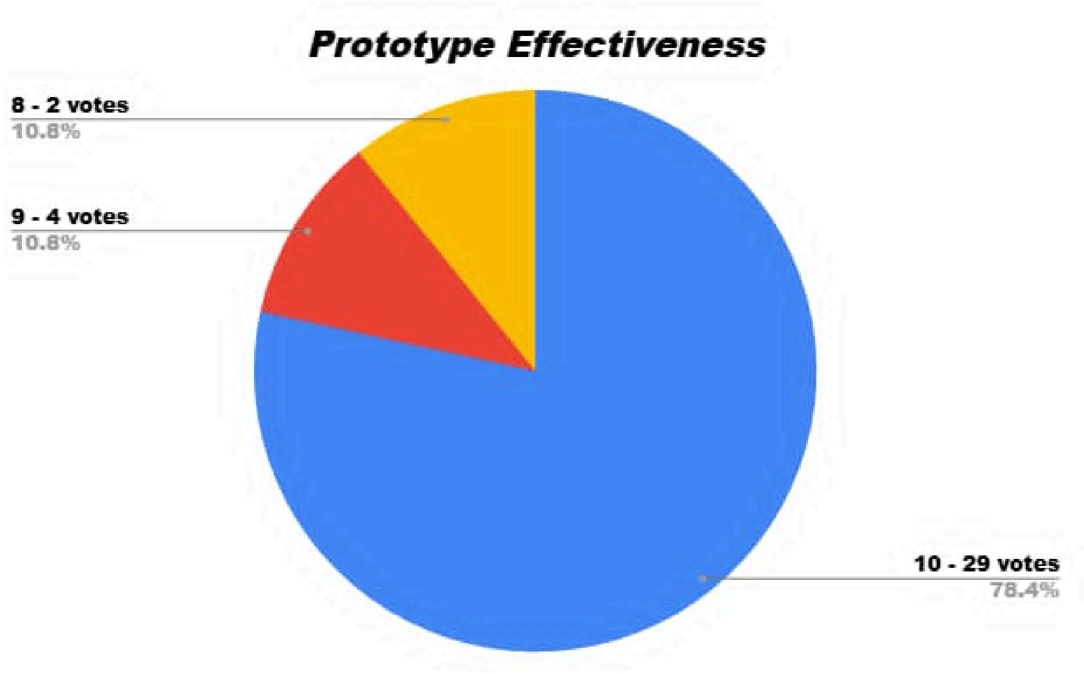


Figure 23. Prototype Effectiveness

The overwhelming majority of evaluators (**82.8% at a rating of 10**) view the VOTECHAIN system as a highly effective solution to electoral discrepancies. By integrating biometric authentication with a decentralized ledger, the system directly mitigates fraud and human error. These results confirm that the technical approach—using blockchain to create an immutable record—effectively addresses the core pain points of traditional voting systems, proving that the innovation is not just technically sound but practically viable for increasing electoral transparency

Qualitative Assessment

In addition to the numerical ratings, qualitative feedback was collected through open-ended responses in the peer evaluation instrument. These responses focused on the description of the prototype, its perceived strengths and weaknesses, and recommendations for improvement. The collected comments were analyzed thematically to identify recurring patterns and insights regarding the system's performance and design.

Prototype Description

Peer evaluators generally described the proposed system as a **technologically advanced electronic voting solution** that integrates embedded hardware with real-time digital processing. The prototype was frequently characterized as **convenient** and **easy to use**, indicating that the system's operational flow was intuitive to first-time users. Several evaluators explicitly noted the presence of **two-factor authentication** and **real-time result generation**, suggesting that the core features of the system were clearly observable during demonstration and use.

Strengths of the System

A major strength identified by the evaluators was the system's **security**. The use of two-factor authentication was frequently mentioned as an effective way to prevent unauthorized voting and reduce the risk of manipulation. In addition, the **real-time display of voting results** was seen as a positive feature that improves transparency and trust in the system.

Evaluators also highlighted the **ease of operation** of the prototype. The interface was generally described as user-friendly, and the overall interaction between the hardware and software components appeared smooth. These observations suggest that the system was able to perform its intended functions effectively during the exhibition.

Weaknesses Identified

Despite the positive feedback, several weaknesses were also noted. The most common concern was the system's **dependence on a stable internet connection**, which may limit its use in areas with poor connectivity. Some evaluators also experienced **delays** during system operation, which affected the overall user experience.

Other concerns included the **cost-effectiveness** of the system and the difficulty of applying the prototype to **large-scale implementations**. A few responses also pointed out the need to further strengthen **software reliability and data handling**, especially for long-term use.

Suggestions for Improvement

Peer evaluators provided several suggestions to improve the prototype. Many recommended using a **larger screen** to make the system easier to use and more accessible. Improvements to the **user interface design** were also suggested, such as clearer layout and the addition of a **loading screen** to inform users during processing. To address performance concerns, some evaluators suggested improving the system's **data-saving speed** and overall processing efficiency. These suggestions indicate possible areas for enhancement in future versions of the system.

The quantitative and qualitative data collected from the 35 peer evaluators indicate that the VOTECHAIN prototype successfully met its objectives. The system received a high overall average rating of **9.75 out of 10**, with the highest marks given for **Demonstration (9.82)** and **Effectiveness (9.77)**. These scores reflect the evaluators' positive reception of the system's ease of use and functionality. While the **Quality** rating was slightly lower at **9.63**, it remains high and aligns with the qualitative feedback regarding specific hardware limitations.

The written comments supported these numerical ratings, highlighting the system's modern design, strong security features, and transparent real-time results as major strengths. However, evaluators also noted the system's reliance on a stable internet connection as a limitation. In conclusion, the evaluation confirms that the prototype is an effective solution for secure voting, though future iterations would benefit from adding offline capabilities to enhance reliability.

CONCLUSION AND RECOMMENDATIONS

Conclusion

This study successfully developed "VOTECHAIN," a secure electronic voting system that integrates two-factor hardware authentication with a custom private blockchain backend to ensure data integrity. The implementation demonstrated that a permissioned ledger architecture utilizing SHA-256 cryptographic chaining and atomic database locks effectively provides tamper-evidence and prevents double-voting exploits without the latency or energy costs of traditional Proof-of-Work systems.

The integration of physical edge devices with a real-time public interface demonstrated a significant advancement in electoral transparency. The system's hardware-software bridge enforced strict voter eligibility through biometric and RFID verification, preventing credential sharing at the source. Furthermore, the project proved that real-time auditing is feasible by using WebSockets to broadcast vote tallies and ledger updates instantly to a public dashboard. Ultimately, the system proved that bridging physical edge devices with a real-time, WebSocket-enabled audit dashboard significantly enhances the transparency and public trust required for modern electoral processes.

Recommendations

While the current iteration of VOTECHAIN meets its core objectives, further improvements are recommended to ensure that the system can be deployed to larger electoral environments.

1. **Transition to a Distributed Network Architecture -** Currently, the system runs on a localized blockchain simulation. To realize true decentralization, future iterations should deploy the blockchain across a mesh of independent nodes

2. **Enhanced Biometric Security** - While the AS608 optical sensor is effective for prototyping, optical technology can be susceptible to spoofing using high-quality latent prints. In the future, the biometric sensor should be upgraded to capacitive fingerprint sensors or integrated vein recognition technology.
3. **Automating the Count with Smart Contracts** - Currently, VOTECHAIN relies on an external Python script to calculate results from a stored JSON file. A more secure approach is to use Smart Contracts which are self-executing code that lives on the blockchain (like Hyperledger Fabric).

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